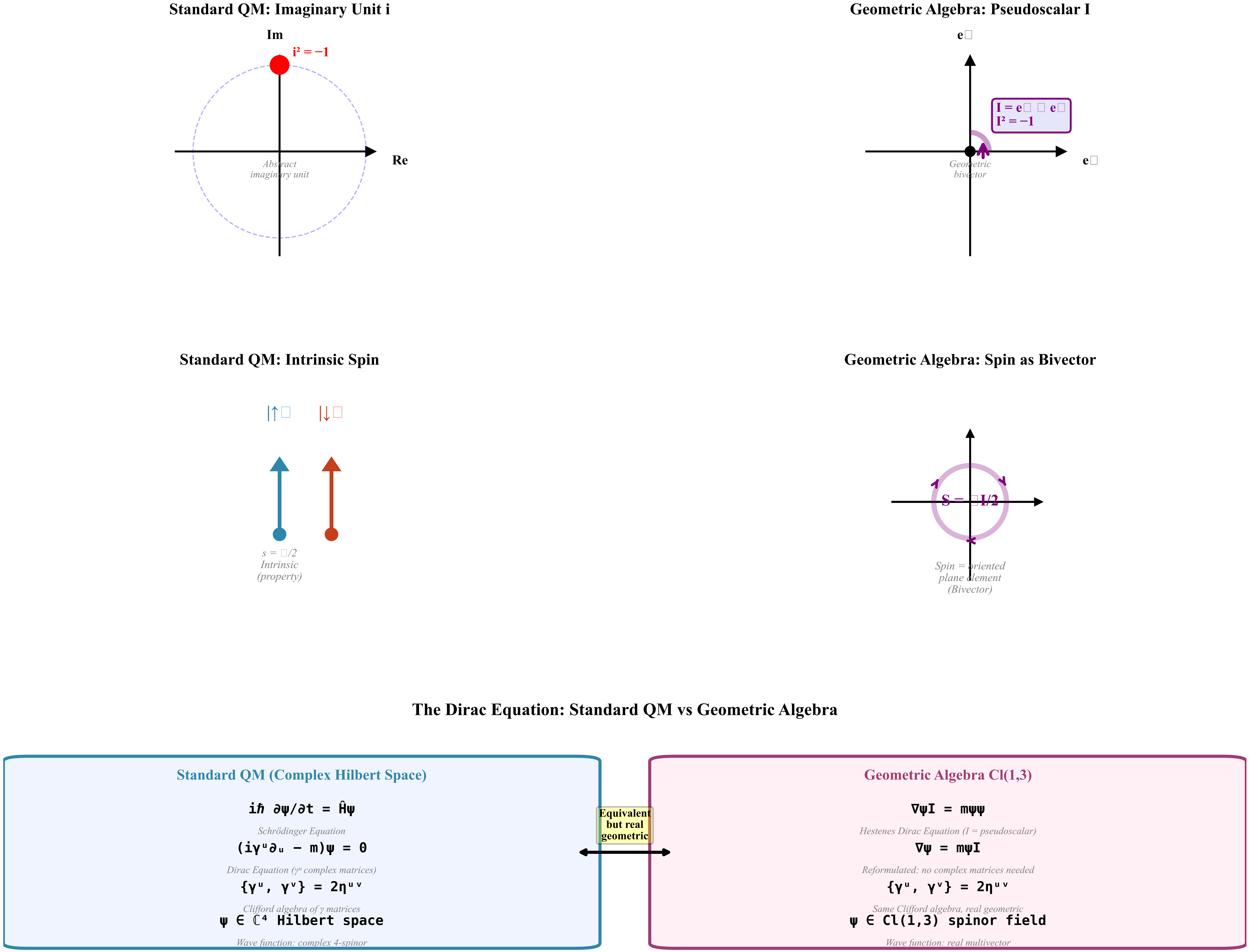

Standard QM: Imaginary Unit i
Im
i² = −1
Re
Abstract imaginary unit
Geometric Algebra: Pseudoscalar I
I² = −1
Geometric bivector
Standard QM: Intrinsic Spin
Intrinsic (property)
Geometric Algebra: Spin as Bivector
Spin = oriented plane element (Bivector)
The Dirac Equation: Standard QM vs Geometric Algebra
Standard QM (Complex Hilbert Space)
iℏ ∂ψ/∂t = Ĥψ
Schrödinger Equation
(iγᵘ∂ᵤ − m)ψ = 0
Dirac Equation (γᵘ complex matrices)
{γᵘ, γᵛ} = 2ηᵘᵛ
Clifford algebra of γ matrices
ψ ∈ ℂ⁴ Hilbert space
Wave function: complex 4-spinor
Equivalent but real geometric
Geometric Algebra Cl(1,3)
∇ψI = mψψ
Hestenes Dirac Equation (I = pseudoscalar)
∇ψ = mψI
Reformulated: no complex matrices needed
{γᵘ, γᵛ} = 2ηᵘᵛ
Same Clifford algebra, real geometric
ψ ∈ Cl(1,3) spinor field
Wave function: real multivector